7. Breadth-First Search (BFS)

Aim:

To implement Breadth-First Search using Python.

Algorithm:

1. Create a graph using an adjacency list.
2. Initialize a queue with the starting node.
3. Remove the first node from the queue and display it.
4. Add its unvisited neighbours to the queue.
5. Repeat until the queue is empty.

Program:

```
from collections import deque
graph = {
'A': ['B', 'C'],
'B': ['D', 'E'],
'C': ['F'],
'D': [],
'E': [],
'F': []
}
def bfs(start):
visited = set()
queue = deque([start])
while queue:
node = queue.popleft()
if node not in visited:
print (node, end=" ")
visited.add(node)
queue. extend (graph[node])
bfs('A')
```

Output:



Result:

Thus, Breadth-First Search was implemented successfully using Python.